

Group Project Instructions

The goal of the course project is to go through the complete data science process to answer questions you have about some topic of your own choosing. You will acquire the data, design your visualizations, run statistical analysis, and communicate the results.

This is a group project where group membership assignments (**2-3 students**) are made by the instructor. I recognize that individual schedules and other constraints might limit your ability to work in a team. If this the case, please let me know **immediately**. In general, I anticipate that all members of a group will receive the same score. However, I reserve the right to assign different scores to each group member based on peer assessments of effort and contribution.

Data Examples

The following are some of the sources we use for data. You can use data there or you can use them for inspiration for project ideas. Using **multiple datasets** could enhance the analysis. Please try to be creative!

- Google Dataset search: <https://datasetsearch.research.google.com/>
- CDC: <https://data.cdc.gov/browse>
- 500 cities: <https://www.cdc.gov/500cities/index.htm>
- UN: <http://data.un.org/>
- Kaggle: <https://www.kaggle.com/datasets>
- AWS: <https://registry.opendata.aws/>
- FEC: <https://www.fec.gov/>
- FiveThirtyEight: <https://github.com/fivethirtyeight/data>

Deadlines for Project Deliverables:

- Project Proposal: April 12th at 11:59pm
- Project Update: May 3rd at 11:59pm
- Final Report and Notebook: May 17th at 11:59pm
- Project Presentation: May 19th before **10:30 am**

Deliverables:

- **Project Proposal:**
Please include:
 - a. Full names of all team members
 - b. One paragraph describing the dataset you plan to use
 - c. One paragraph outlining your goals for the project. The specific question(s) you aim to answer should be clear
 - d. The source(s) of your data

Your project proposal is a chance to get early feedback on your ideas. While the proposal document is short, your proposal should be well thought out and clear.

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- **Project Update:** maximum of 2 pages explaining current progress. The update file should also have the full names of the students in the group. This document is a way to receive feedback on your progress before completing the final report and presentation. Please be aware that I will only see what you write about in the update, so if you want detailed feedback on your work, you should provide detailed information in your update. Anything that you are having issues with or need advice on should be written about in the update as well.
- **Presentation:** Each group will present their findings to the rest of the class during the last week(s) of class. The presentation will be a 5 min **lightning talk** using slides. Each team member must present at least one slide (not including the title slide). You will be cut-off at the 5-minute mark so *please* practice with a timer before your assigned presentation day. There are many good online guides for creating a good presentation. Here is [one](#).
- **Report:** The report should reflect comments and feedback received during the proposal and update. **The reports should be at most five pages long.** This report should stand alone, and I should not have to refer to your notebook to understand what you have written or to see any figures or tables. See more about the report in the next sections.
- **Notebook:** a high-quality, well-organized, and readable Python Jupyter Notebook. You should make use of both markdown and code cells and document your code. You should also think about things such as code quality, reusability, and replicability.

Report Objectives

The final report should cover these aspects:

- **Overview and Motivation:** Provide an overview of the project goals and the motivation for it.
- **Related Work:** Anything that inspired you, such as a paper, a newspaper/magazine article etc.
- **Initial Questions:** What questions are you trying to answer? How did these questions evolve over the course of the project? What new questions did you consider during your analysis?
- **Data:** Source, scraping method, cleanup, etc.
- **Exploratory Analysis:** What visualizations did you use to look at your data in different ways? What are the different statistical methods you considered? Justify the decisions you made. How did you reach these conclusions?
- **Final Analysis:** What did you learn about the data? How did you answer the questions? How can you justify your answers?
- **Group member contributions:** Please state at the end the role each member of the team had in the project.

Report Quality

Similarly to the project in DATA 11800, the quality of the writing, tables and figures is very important. Make sure that **the report does not exceed 5 pages** (not including references).

Some important things to keep in mind:

- If you are referencing someone else's work (which you should be!) whether it is a journal article, blog post, or news story, you need to provide a citation. **A URL is not sufficient and is not considered a**

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citation. You may use any citation style you like as long as you include both in-text citations and a list of references at the end of your report. See [this guide](#) from the UChicago Library for more information about formatting citations.

- Figures should meet the following standards:
 - Must use the correct type of graph for the data being depicted (i.e. bar charts for count data, line graphs for longitudinal data, etc.)
 - Must be clearly labeled with figure numbers and referenced by those labels in the text of your report
 - When referring to the figure in text the word ‘Figure’ should be capitalized (eg Figure 1 not figure 1).
 - Axes should be labeled and informative, including any necessary units etc. They should never be variable names from your Python DataFrame (the default in `matplotlib`). For example, instead of “daily_temp”, you would write “Daily Temperature (Celsius)” or something similar.
 - Try to only include the most relevant plots (10 of the same kind of plot of a slight variation of the same information isn’t very interesting).
 - Do not include too much information in a single plot (for example, if you compute some covid or crime metric for each community area in Chicago, you should not display that metric in one giant barplot with all community areas listed. Points will be taken off for things like this).
 - Be creative! I love to see good visualizations.
- Similarly, tables should meet the following standards:
 - Must have numbers, captions, text references, be placed near their reference, and “Table” should be capitalized similar to figures.
 - Should NOT be screenshots of a `group_by` in Python. They should be well-formatted, with no variable names or underscores (like figure axis labels).
 - Any decimal numbers in table cells should be rounded to some reasonable decimal place for ease of reading and interpretation.
- In general, Figures and Tables should make the information easier for the reader to digest than if you were to write that information out in text. If you make a figure or table and notice that you need to look at it for over 30-60s to understand what it is saying, you may need to make edits.
- You should not include screenshots of code in your report, or any code in general. If we want to see your code, we can go to your notebook, but it should not be in your report.
- Take this time to practice interpreting results from the models you are learning. Instead of listing coefficients or metrics, explain what they mean. In general, try to synthesize information into a coherent argument as opposed to listing things and moving on to the next section.
- Make sure you take time to really discuss the broader implications of your work as well as limitations and anything that you would do if you had more time.